

Chapter 23

Using Scene Detection

If you have a program that someone has delivered as a single media file, with no accompanying EDL with which to split it up, you can use DaVinci Resolve's Scene Detect window to automatically find the cut points and split it into individual clips, ready for grading.

Contents

Scene Cut Detection on the Timeline (Studio Version Only)	458
Scene Detection in the Media Page	459
The Scene Detect Window Interface	459
The Scene Detect Viewers	459
The Scene Detect Graph	461
Cut List	462
The Scene Detect Options Drop-down Menu	463
An Example Scene Detect Workflow	463

Scene Cut Detection on the Timeline (Studio Version Only)

If you need to break down a previously edited video into its component clips for re-editing or color correction, you can now do so directly in an Edit or Cut page Timeline. Using the DaVinci Neural Engine, DaVinci Resolve can automatically analyze and split up an edited video into individual clips.

If you prefer, you can continue to use the original Scene Cut Detection tool found in the Media Pool and described later in this chapter.

To use Scene Cut Detection on the Timeline:

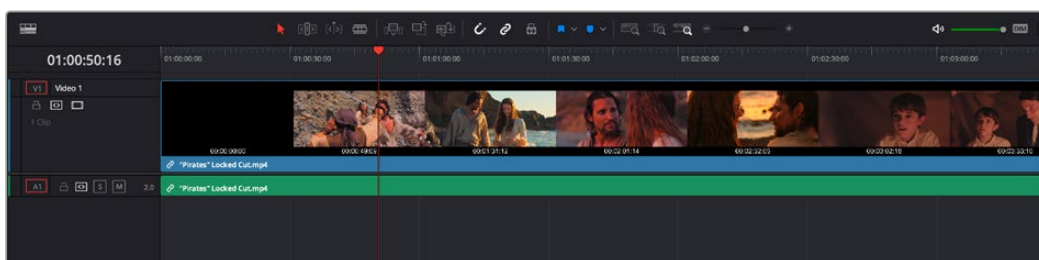
- 1 Put one or more clips you want to split on the Timeline. If you have clips on more than one video track, you can selectively lock/unlock tracks or enable/disable the Auto Track Selectors to limit the scene detection to specific video tracks. Additionally, you can limit Scene Cut Detection to just a portion of a clip by setting In and Out points on the Timeline around the section you want to analyze.
- 2 Choose Timeline > Detect Scene Cuts.

A dialogue box appears, “Detecting scene cuts in clips x of x.” This process can take some time, depending on the length, number, and complexity of the clips you’ve selected. When the Scene Cut Detection has finished, the clip you selected will be broken up into a number of through edits that now can be used as independent clips.

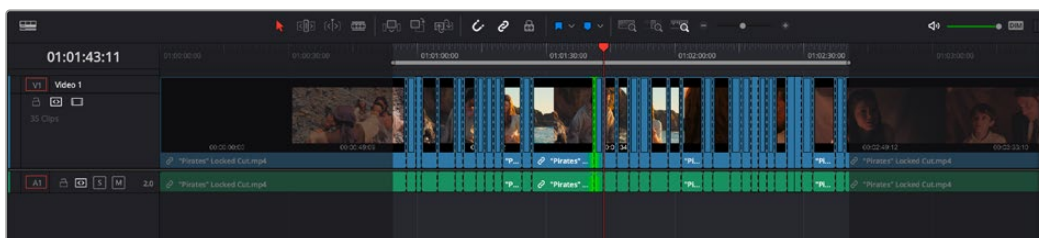
Checking and Fixing Your Results

If the Neural Engine has made an error, you can fix it manually by navigating to the cut using the Up and Down Arrow keys to go back and forth in the Timeline, and by then doing one of the following:

- **To remove a Cut:** Click the through edit to select it, and press the “Delete” key.
- **To make a New Cut:** Place the timeline indicator at the cut point, and choose Timeline > Split Clips (Command-I).



A single clip of a finished edit, consisting of multiple cuts before the Detect Scene Cuts command



Multiple individual clips extracted from the edited clip via Detect Scene Cut; the operation has been contained by the In and Out points, and one of the resulting through edits has been highlighted in green.

Scene Detection in the Media Page

You can still perform automatic scene detections using the original tools in the Media page. When using the original Media page Scene Detection, it's important to note that the analysis and splitting of the selected clip is done in the Media Storage panel, before the clip is added to the Media Pool.

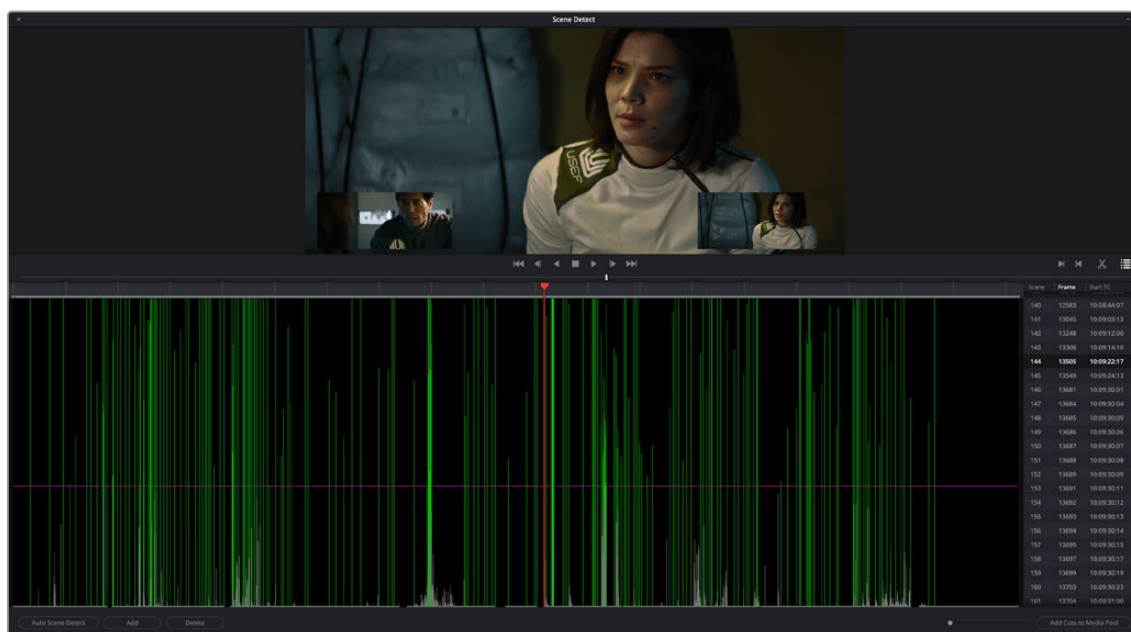
To open a clip into the Scene Detect window:

- 1 Open the Media page, and use the Media Storage browser to find and select the clip you need to split apart. Do not add a clip you want to use scene detection on to the Media Pool first. You need to use Scene Detection before the clip has been imported.
- 2 Right-click the file and choose Scene Cut Detection.
- 3 The Scene Detect window opens up, containing the clip you selected.
- 4 Press the Auto Scene Detect button in the lower left of the interface.

DaVinci Resolve scans through the selected scene and analyzes the media for possible cut points.

The Scene Detect Window Interface

The Scene Detect window is divided into three main areas, the viewers, the Graph, and the Cut List. Together, these controls let you analyze the movie, examine the automatically found cuts, and manage the Cut List in preparation for sending back to your project.



Scene Detect window

The Scene Detect Viewers

A set of three viewers appear at the top of the Scene Detect window, one main viewer and two smaller picture-in-picture (PiP) viewers. These three viewers are designed to make it easy to test whether the playhead in the Scene Detect Graph is on a cut point or not. The leftmost PiP viewer is the last outgoing frame of a detected cut point. The main viewer shows the first incoming frame of that cut point, and the rightmost PiP viewer shows the second incoming frame of that cut point.

If the playhead in the Scene Detect Graph is directly on top of an edit point, the leftmost viewer should show a completely different frame than the center and rightmost viewers, which should be very similar to one another. If this is the case, you have found a correct cut point. This can be seen in the following example.



The Scene Detect viewers show the last frame of the outgoing clip, and the first two frames of the incoming clip. If the left viewer is significantly different than the main and right viewers, this indicates a correct cut point.

If all three viewers appear to display a continuous series of frames, then you're not looking at a cut point.



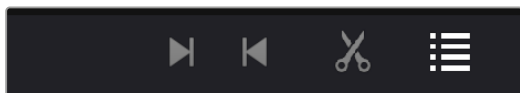
No scene cut here as all images are almost the same

Underneath the viewers are a series of controls.



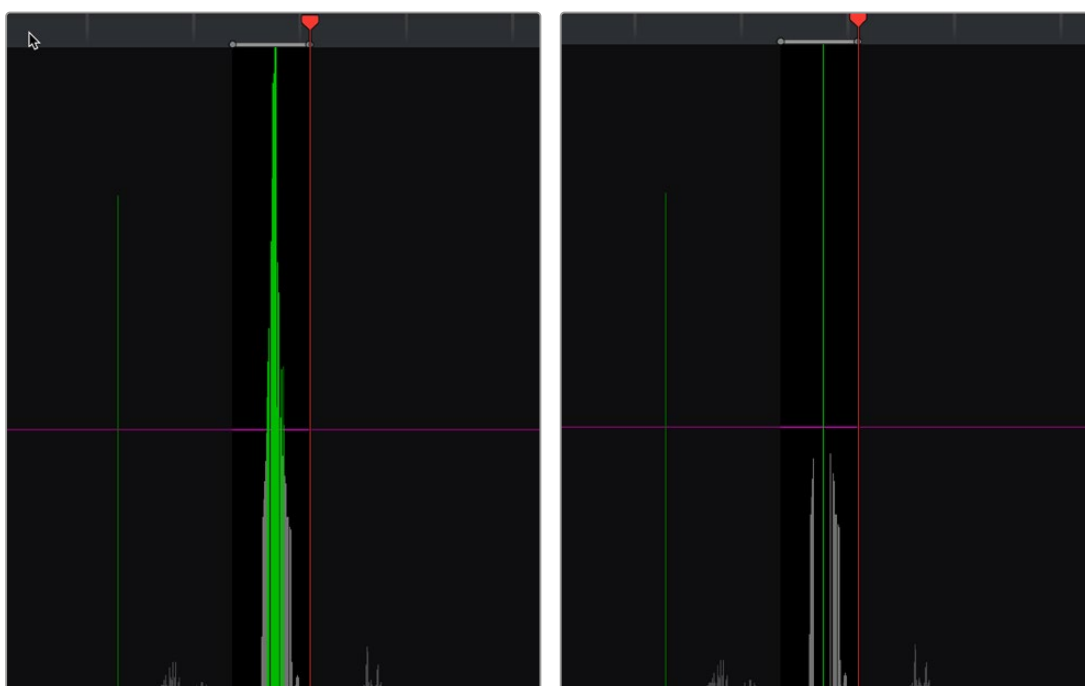
The Scene Detect viewer transport controls

- **Transport controls:** A set of seven transport controls include first frame, step back, play reverse, stop, play forward, step forward, and last frame.



The In, Out, Prune, and Show Cut List controls

- **In:** Lets you set an In point, with which to define a range of the Scene Detection Graph to prune.
- **Out:** Lets you set an Out point, with which to define a range of the Scene Detection Graph to prune.
- **Prune:** If you've identified a large number of false positive scene cuts (for example, a cluster of cuts corresponding to a dissolve from one shot to another), use the In and Out buttons to surround the undesirable range of scene cuts in the Scene Detect Graph, and then click Prune to eliminate all scene cuts between these points that are within one frame of another scene cut. Within the group of identified cuts, the highest probability cut will remain while the other cuts are deleted.



(Left) Isolating scene cuts to prune with In and Out points, (Right) The result of clicking the Prune button to eliminate all unwanted scene cuts but one

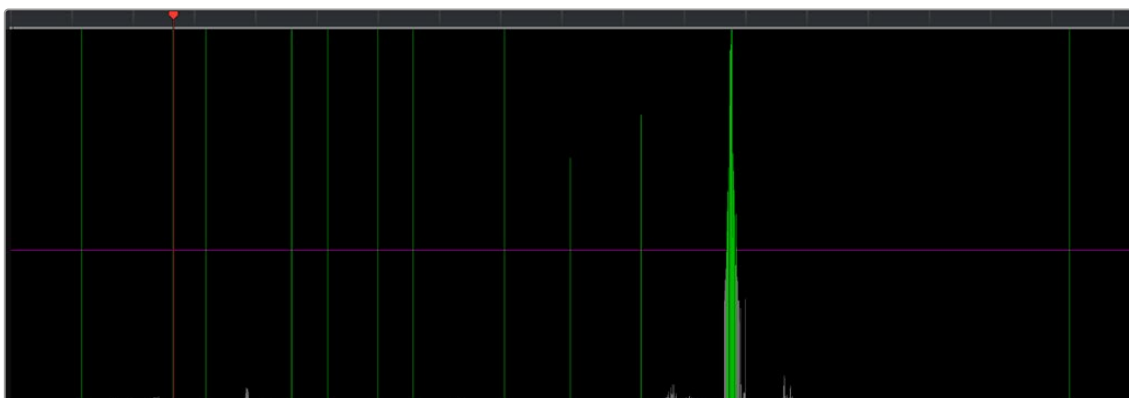
- **Show Cut List:** Shows and hides the Cut List, which shows the currently detected scene cuts.

The Scene Detect Graph

The majority of the bottom half of the Scene Detect window, to the left, consists of the Scene Detect Graph, which shows the scene detect analysis results after you've clicked the Start button.

Frames that DaVinci Resolve thinks are cut points appear as green vertical “scene cuts” of various heights. The height of each scene cut corresponds to the likelihood that frame is really an edit point, and not a swish pan, sudden jump in the motion of the frame, or abrupt change in color or lighting, all of which can fool the scene detection algorithm.

A horizontal magenta confidence bar lets you choose the threshold of confidence required for scene cuts to be added to the Cut List. If you drag this bar up above any shorter scene cuts of low confidence, those lines turn gray and are omitted from the Cut List.



Detection graph displays potential scene cuts

NOTE: Dissolves and other transitions are not automatically detected, although dissolves most often appear as a triangular cluster of lines peaking in the middle.

Four controls appear underneath the graph.

- **Auto Scene Detect:** This initiates the scene cut detection process.
- **Add:** Lets you manually add a scene cut at the current position of the playhead. Sometimes two adjacent clips with similar color and lighting will appear to be a single clip to the scene detection algorithm. This lets you add scene cuts at frames where they weren't initially found.
- **Delete:** Lets you manually delete a scene cut located at the position of the current frame indicator within the graph.
- **Zoom slider:** Lets you zoom into and out of the Scene Detect Graph to see more or less detail as you examine the results.

Cut List

At the lower right of the Scene Detect window, the Cut List displays one entry for each of the scene cuts that intersect the confidence bar.

Scene	Frame	Start TC
0	0	00:00:00:00
1	131	00:00:05:11
2	300	00:00:12:12
3	361	00:00:15:01
4	520	00:00:21:16
5	587	00:00:24:11
6	679	00:00:28:07
7	745	00:00:31:01
8	914	00:00:38:02
9	1036	00:00:43:04
10	1167	00:00:48:15
11	1325	00:00:55:05
12	1326	00:00:55:06
13	1327	00:00:55:07
14	1328	00:00:55:08
15	1329	00:00:55:09
16	1331	00:00:55:11
17	1332	00:00:55:12
18	1333	00:00:55:13
19	1334	00:00:55:14
20	1335	00:00:55:15
21	1336	00:00:55:16
22	1337	00:00:55:17

Add Cuts to Media Pool

The Cut List shows all currently detected cuts

Three columns show each cut's order number, frame number, and timecode value. You can select items in the Cut List to evaluate each cut using the three viewers above. Whenever you select a new item in the Cut List, the playhead jumps to that frame in the Scene Detect Graph.

To select items in the Cut List:

- Click any item in the Cut List.
- Press N (next) or the Down Arrow to select the next item down.
- Press P (previous) or the Up Arrow to select the next item previous.

As you move up and down the list, you can delete items that you can confirm aren't real cuts using the viewers above. If it's a long list and you don't have time to check it all at once, it can be saved for later recall using commands found in the Scene Detect Options drop-down menu.

Once you're finished checking the list and are satisfied that each cut is accurate, you can use it to split the media file into individual clips in the Media Pool by clicking "Add Cuts to Media Pool," located immediately below.

The Scene Detect Options Drop-down Menu

The Options drop-down menu, located at the upper right-hand corner of the Scene Detect window, contains a variety of commands.

- **Reset Zoom:** Sets the zoom level of the Scene Cut Graph such that the entire clip fits within the current width.
- **Reset Marks:** Clears the current In and Out points you've set.
- **Prune Scene Cuts:** If you've identified a large number of false positive scene cuts (for example, a cluster of cuts corresponding to a dissolve from one shot to another), use the In and Out buttons to surround the undesirable range of scene cuts in the Scene Detect Graph, and then click Prune Scene Cuts to eliminate all scene cuts between these points that are within one frame of another scene cut. Within the group of identified cuts, the highest probability cut will remain while the other cuts are deleted.
- **Save Scene Cut:** Saves the current scene cut detection information, including the probability metadata, to disk. Scene Cut files use the file extension .sc and can be reimported later to continue working on a lengthy scene detection task.
- **Load Scene Cut:** Imports an existing .sc file into the Scene Detect window. You must first open the media file you're working on into the Scene Detect window before you can load a Scene Cut file.
- **Save EDL:** Exports the Cut List as a CMX-style EDL.
- **Load EDL:** Loads a CMX-style EDL into the Cut List, letting you use the cut information from the EDL during the Scene Cut Detection process.
- **Auto Cue:** When enabled, the playhead jumps to each new scene cut as it's detected when you initiate scene detection. This lets you evaluate each scene cut as it's found using the three viewers above.

An Example Scene Detect Workflow

This section describes an ideal workflow for using scene detection without an EDL.

To scene detect a media file:

- 1 Locate a media file to scene detect using the Media Storage browser of the Media page.
- 2 Verify its frame rate and if it uses drop-frame timecode, and make sure that the "Timeline frame rate" matches the "Use drop frame timecode" parameter in the Master Settings panel of the Project Settings. These parameters are not automatically set if the project already has media in the Media Pool, and you may have problems if they don't match your media.
- 3 Right-click the media file, and choose Scene Cut Detection.

- 4 When the Scene Detect window appears, click the Options drop-down menu and choose Auto Cue (it should be on by default, but it's always good to check), then click Auto Scene Detect.

Scene detection initiates, and you can evaluate each scene cut as it's found. If any scene cut looks wrong (three sequential frames in a row), note its place in the list for future evaluation.

- 5 When DaVinci Resolve has finished scene detection, move the playhead to some of the shorter scene cuts, and verify if they're actual cuts by checking the three viewers above. If the frames being displayed are "different-same-same," then it's a genuine cut.

If the frames being displayed are "same-same-same" (actually three sequential frames), then these aren't cuts.

TIP: Fast camera motion such as whip pans, sudden changes in lightness such as camera flashes, or even film coming up to speed causing the shutter to “flash” can confuse the analysis, which looks for large changes in the image.

- 6 If there are numerous low-confidence scene cuts that you’ve verified aren’t cuts, drag the magenta confidence bar so that the low-confidence scene cuts fall below it to automatically remove them all from the list.
- 7 Next, you may want to move down the Cut List, evaluating each scene cut to verify that it’s correct. Click the first scene cut in the list, check it, then press the keyboard Down Arrow key to select the next list item down, check it, and repeat until you’ve checked every item in the list. If you need to move back up the list, you can press the Up Arrow key to select the previous list item. If any item is not a cut point, click the “Delete” button at the bottom left corner of the Scene Detect window to eliminate that scene cut.
- 8 If there are sections in the Scene Detect Graph with dense groups of spikes, these are probably frames with types of motion that confused the Scene Cut Detector. To delete this unwanted “noise” in the data, use the In and Out buttons to isolate the data, and then click “Prune” to delete these unwanted scene cuts.
- 9 If there’s a gap between any two scene cuts that you’re positive should have another scene cut, then scrub the playhead or use the transport controls to find the missing cut, and click the “Add” button at the bottom left corner of the Scene Detect window to add another scene cut.

TIP: Adjacent shots with very similar ranges of color and contrast may sometimes go undetected by the scene detection algorithm. If you know of scenes in the media you’re analyzing that are like this, you may want to scrub through them a bit more carefully to make sure you’re not missing anything. However, if you find you’ve missed a cut later, you can always use the Split Clip control in the Edit page timeline to add a new edit point.

- 10 When you’re confident that the Cut List is accurate, split the media file into individual clips in the Media Pool by clicking “Add Cuts to Media Pool.”
- 11 When the Conform Settings dialog appears, click OK if you checked your settings in step 2.
- 12 Close the Scene Detect window.

The individually cut up clips of the media file you analyzed now appear in the Media Pool, and you can edit the entire sequence of clips into a new Timeline in order, ready for grading.